

Dualizability in Symmetric Monoidal Category

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This is devoted for the categorical details in

We'll follow the notations Aditya covered in the talk. Fix $(\mathcal{C}, \otimes, I)$ as our symmetric monoidal category.

1 Weak Dualizability

Definition 1.1. Given an object $A \in \mathcal{C}$, an object $B \in \mathcal{C}$ is called A 's *weak dual*, if there exists a natural isomorphism $\xi : \mathcal{C}(- \otimes A, I) \xrightarrow{\sim} \mathcal{C}(-, B)$ between functors $\mathcal{C}^{\text{op}} \rightarrow \text{Set}$. In this case, A is said to be *weakly dualizable*.

One important fact one needs to verify is the uniqueness (up to isomorphism), this is indeed true:

Proposition 1.2. *If $A \in \mathcal{C}$ is weakly dualizable, its weak dual is unique up to isomorphism.*

Proof. This is a Yoneda Lemma type argument. Let $B, B' \in \mathcal{C}$ be two weak duals of A , then there is a natural isomorphism $\xi : \mathcal{C}(-, B) \xrightarrow{\sim} \mathcal{C}(- \otimes A, I) \xrightarrow{\sim} \mathcal{C}(-, B')$ (abuse of notation, still use ξ as natural isomorphism).

Using the isomorphisms $\xi_B : \mathcal{C}(B, B) \xrightarrow{\sim} \mathcal{C}(B, B')$ and $\xi_{B'}^{-1} : \mathcal{C}(B', B') \xrightarrow{\sim} \mathcal{C}(B', B)$, we get the following two morphisms:

$$\eta := \xi_B(\text{id}_B) : B \rightarrow B', \quad \mu := \xi_{B'}^{-1}(\text{id}_{B'}) : B' \rightarrow B$$

Then, consider the following two diagrams using the naturality:

$$\begin{array}{ccc} \mathcal{C}(B, B) & \xrightarrow[\sim]{\xi_B} & \mathcal{C}(B, B') \\ \downarrow - \circ \mu & & \downarrow - \circ \mu \\ \mathcal{C}(B', B') & \xrightarrow[\sim]{\xi_{B'}} & \mathcal{C}(B', B') \end{array} \quad \begin{array}{ccc} \text{id}_B & \xrightarrow{\quad} & \eta \\ \downarrow & & \downarrow \\ \mu & \xrightarrow{\quad} & \text{id}_{B'} = \eta \circ \mu \end{array}$$

$$\begin{array}{ccc} \mathcal{C}(B', B') & \xrightarrow[\sim]{\xi_{B'}^{-1}} & \mathcal{C}(B', B) \\ \downarrow - \circ \eta & & \downarrow - \circ \eta \\ \mathcal{C}(B, B') & \xrightarrow[\sim]{\xi_B} & \mathcal{C}(B, B) \end{array} \quad \begin{array}{ccc} \text{id}_{B'} & \xrightarrow{\quad} & \mu \\ \downarrow & & \downarrow \\ \eta & \xrightarrow{\quad} & \text{id}_B = \mu \circ \eta \end{array}$$

This shows μ, η are mutual inverses, hence $B \cong B'$. $\square$

If the weakly dualizable objects are small, then there's no ambiguity choosing a representative of A , denote as DA .

Another notion is how to “formally evaluate” weakly dualizable objects:

Definition 1.3. Given $A \in \mathcal{C}$ that's weakly dualizable, consider the isomorphism

$$\xi_{DA} : \mathcal{C}(DA \otimes A, I) \xrightarrow{\sim} \mathcal{C}(DA, DA)$$

Define the *evaluation morphism* $\epsilon_A : DA \otimes A \rightarrow I$ as the dual to the following identity map, $\epsilon_A := \xi_{DA}^{-1}(\text{id}_{DA})$.

2 Functoriality of Dualization

Now, let $\mathcal{C}^*$ denote the full subcategory of all weakly dualizable objects, the aim is to show dualization is functorial, i.e., there is a functor $D : (\mathcal{C}^*)^{\text{op}} \rightarrow \mathbf{Set}$. It's clear that DA is well-defined (up to isomorphism), so the morphism (and preservation of composition) is the actual problem.

Definition 2.1. Given a morphism $f : A \rightarrow A'$ in $\mathcal{C}^*$, consider the morphism

$$DA' \otimes A \xrightarrow{\text{id}_{DA'} \otimes f} DA' \otimes A' \xrightarrow{\epsilon_{A'}} I$$

Using the isomorphism $\xi_{DA'} : \mathcal{C}(DA' \otimes A, I) \xrightarrow{\sim} \mathcal{C}(DA', DA)$, define

$$Df := \xi_{DA'}(\epsilon_{A'} \circ (\text{id}_{DA'} \otimes f)) : DA' \rightarrow DA$$

The goal is to show this dualization of morphism is functorial. Before that, fix $f : A \rightarrow A'$ in $\mathcal{C}^*$, and $Df : DA' \rightarrow DA$ in $\mathcal{C}$, consider the following commutative diagram using the natural isomorphism $\xi : \mathcal{C}(- \otimes A, I) \xrightarrow{\sim} \mathcal{C}(-, DA)$:

$$\begin{array}{ccc} \mathcal{C}(DA \otimes A, I) & \xrightarrow[\sim]{\xi_{DA}} & \mathcal{C}(DA, DA) \\ \downarrow \scriptstyle - \circ (Df \otimes \text{id}_A) & & \downarrow \scriptstyle - \circ Df \\ \mathcal{C}(DA' \otimes A, I) & \xrightarrow[\sim]{\xi_{DA'}} & \mathcal{C}(DA', DA) \end{array} \quad \begin{array}{ccc} \epsilon_A & \dashv & \text{id}_{DA} \\ \downarrow & & \downarrow \\ \epsilon_A \circ (Df \otimes \text{id}_A) & \dashv & Df \end{array}$$

So, we see the equality $\xi_{DA'}(\epsilon_A \circ (Df \otimes \text{id}_A)) = Df$, and don't forget that $Df := \xi_{DA'}(\epsilon_{A'} \circ (\text{id}_{DA'} \otimes f))$. So, the isomorphism deduces $\epsilon_A \circ (Df \otimes \text{id}_A) = \epsilon_{A'} \circ (\text{id}_{DA'} \otimes f)$.

Using this, we deduce the functoriality as morphism $DA'' \otimes A \rightarrow I$:

Lemma 2.2. $D : (\mathcal{C}^*)^{\text{op}} \rightarrow \mathbf{Set}$ is functorial, namely $D(g \circ f) = Df \circ Dg$.

Proof. Given $f : A \rightarrow A'$ and $g : A' \rightarrow A''$ in $\mathcal{C}^*$, we get the following equality:

$$\begin{aligned} \epsilon_{A''} \circ (\text{id}_{DA''} \otimes (g \circ f)) &= \epsilon_{A''} \circ (\text{id}_{DA''} \otimes g) \circ (\text{id}_{DA''} \otimes f) = \epsilon_{A'} \circ (Dg \otimes \text{id}_{A'}) \circ (\text{id}_{DA''} \otimes f) \\ &= \epsilon_{A'} \circ (Dg \otimes f) = \epsilon_{A'} \circ (\text{id}_{DA'} \otimes f) \circ (Dg \otimes \text{id}_A) \\ &= \epsilon_A \circ (Df \otimes \text{id}_A) \circ (Dg \otimes \text{id}_A) = \epsilon_A \circ ((Df \circ Dg) \otimes \text{id}_A) \end{aligned}$$

Hence, apply $\xi_{DA''}$, we get:

$$Df \circ Dg = \xi_{DA''}(\epsilon_A \circ (Df \circ Dg) \otimes \text{id}_A) = \xi_{DA''}(\epsilon_{A''} \circ (\text{id}_{DA''} \otimes (g \circ f))) = D(g \circ f)$$

If one is interested they can verify the above are all well-defined :) □

3 Reflexivity and Strong Dualizability

For this part, we'll temporarily assume the chosen object $A \in \mathcal{C}$ is weakly dualizable, and its dual $DA \in \mathcal{C}$ is also weakly dualizable (so DDA exists). Not sure if this assumption is required, but I suspect in a screwed enough symmetric monoidal category, double dual could fail to exist.

Definition 3.1. Consider the isomorphism

$$\mathcal{C}(DA, DA) \xrightarrow{\xi_{DA}^{-1}} \mathcal{C}(DA \otimes A, I) \cong \mathcal{C}(A \otimes DA, I) \xrightarrow{\xi_A} \mathcal{C}(A, DDA)$$

Define $\delta_A : A \rightarrow DDA$ as the image of id_{DA} under this map.

With this in mind, we have the following definition:

Definition 3.2. An object $A \in \mathcal{C}$ is *reflexive*, if DDA exists, and $\delta_A : A \rightarrow DDA$ is an isomorphism.

Now, suppose further that $A, B, A \otimes B$ are all weakly dualizable (again, I don't think $\mathcal{C}^*$ is closed under monoidal product in general, so we need to be careful):

Definition 3.3. Consider the following two (identical) morphisms:

$$\begin{aligned} (DA \otimes A) \otimes (DB \otimes B) &\cong (DA \otimes A) \otimes (DB \otimes B) \xrightarrow{\text{id}_{DA \otimes A} \otimes \epsilon_B} (DA \otimes A) \otimes I \xrightarrow{\epsilon_A \otimes \text{id}_I} I \otimes I \cong I \\ (DA \otimes A) \otimes (DB \otimes B) &\cong (DA \otimes A) \otimes (DB \otimes B) \xrightarrow{\epsilon_A \otimes \text{id}_{DB \otimes B}} I \otimes (DB \otimes B) \xrightarrow{\text{id}_I \otimes \epsilon_B} I \otimes I \cong I \end{aligned}$$

Under the isomorphism

$$\mathcal{C}((DA \otimes DB) \otimes (A \otimes B), I) \xrightarrow{\xi_{DA \otimes DB}} \mathcal{C}(DA \otimes DB, D(A \otimes B))$$

the proposed morphism maps to a morphism $\mu_{A,B} : DA \otimes DB \rightarrow D(A \otimes B)$.

Proposition 3.4. *The unit I is self dual: DI exists, and $I \cong DI$.*

Proof. Using the isomorphism $\mathcal{C}(- \otimes I, I) \cong \mathcal{C}(-, I)$ by the unit axiom of symmetric monoidal category, we see $I \cong DI$ using the uniqueness (up to isomorphism) of weak dual object. □

Definition 3.5. Given $A \in \mathcal{C}$ such that $DA, DDA, D(A \otimes DA)$ all exists, then A is *strongly dualizable*, if A is reflexive, and the morphism

$$DA \otimes A \xrightarrow{\text{id}_{DA} \otimes \delta_A} DA \otimes DDA \xrightarrow{\mu_{A, DA}} D(A \otimes DA)$$

is an isomorphism, equivalently, $A \otimes DA$ is self-dual via $\mu_{A,DA} : DA \otimes DDA \rightarrow D(A \otimes DA)$. Which, DA is also called a *strong dual* of A (and vice versa).

In this case, the *coevaluation morphism* $\eta_A : I \rightarrow A \otimes DA$ is the composite of the following morphism:

$$I \cong DI \xrightarrow{D\epsilon_A} D(A \otimes DA) \xrightarrow{\mu_{A,DA}^{-1}} DA \otimes DDA \xrightarrow{\text{id}_{DA} \otimes \delta_A^{-1}} DA \otimes A \cong A \otimes DA$$

Theorem 3.6. *Given $A, B \in \mathcal{C}$ and the existence of a morphism $\epsilon : A \otimes B \rightarrow I$, TFAE:*

(1) *B is a strong dual of A , and ϵ is the evaluation morphism.*

(2) *There exists a morphism $\eta : I \rightarrow A \otimes B$, such that*

$$\begin{aligned} A &\cong I \otimes A \xrightarrow{\eta \otimes \text{id}_A} A \otimes B \otimes A \xrightarrow{\text{id}_A \otimes \epsilon} A \otimes I \cong A \\ B &\cong B \otimes I \xrightarrow{\text{id}_B \otimes \eta} B \otimes A \otimes B \xrightarrow{\epsilon \otimes \text{id}_B} I \otimes B \cong B \end{aligned}$$

are identities.

(3) *Given any $X, Y \in \mathcal{C}$, the set map*

$$\phi_{XY} : \mathcal{C}(X, Y \otimes B) \rightarrow \mathcal{C}(X \otimes A, Y), \quad (X \xrightarrow{f} Y \otimes B) \mapsto (X \otimes A \xrightarrow{f \otimes \text{id}_A} Y \otimes B \otimes A \xrightarrow{\text{id}_Y \otimes \epsilon} Y \otimes I \cong Y)$$

is an isomorphism. (So, $((- \otimes A) \dashv (- \otimes B))$ are adjoint pairs).

Proof.

(1) $\implies$ (2):

(2) $\implies$ (3): Suppose (2), we ought to construct an inverse of ϕ_{XY} . Given $X \otimes A \xrightarrow{g} Y$, consider the following morphism

$$X \cong X \otimes I \xrightarrow{\text{id}_X \otimes \eta} X \otimes A \otimes B \xrightarrow{g \otimes \text{id}_B} Y \otimes B$$

This defines a map $\psi_{XY} : \mathcal{C}(X \otimes A, Y) \rightarrow \mathcal{C}(X, Y \otimes B)$. To show they're inverse equivalence, first consider arbitrary $f : X \rightarrow Y \otimes B$, we get:

$$\psi_{XY} \circ \phi_{XY}(f) : X \cong X \otimes I \xrightarrow{\text{id}_X \otimes \eta} X \otimes A \otimes B \xrightarrow{f \otimes \text{id}_{A \otimes B}} Y \otimes B \otimes A \otimes B \xrightarrow{\text{id}_Y \otimes \epsilon \otimes \text{id}_B} Y \otimes I \otimes B \cong Y \otimes B$$

This is equal to the following:

$$X \otimes I \xrightarrow{f \otimes \text{id}_I} Y \otimes B \otimes I \xrightarrow{\text{id}_Y \otimes \text{id}_B \otimes \eta} Y \otimes B \otimes A \otimes B \xrightarrow{\text{id}_Y \otimes \epsilon \otimes \text{id}_B} Y \otimes I \otimes B \cong Y \otimes B$$

Using the fact $(\eta \otimes \text{id}_B) \circ (\text{id}_B \otimes \eta) \cong \text{id}_B$ (after suitable isomorphism), tensoring with Y the above reduces to

$$X \otimes I \xrightarrow{f \otimes \text{id}_I} Y \otimes B \xrightarrow{\text{id}_Y \otimes \text{id}_B} Y \otimes B$$

Apply similar proofs to the dual checks out the isomorphism (yeah I'm tired of tracing through it :)

(3) $\implies$ (1): Fix $Y := I$, this induces a natural isomorphism $\mathcal{C}(X, B) \cong \mathcal{C}(X, I \otimes B) \cong \mathcal{C}(X \otimes A, I)$, so B is a weak dual of A ; fix Y

□